

TITLE:

Prescribing trends analysis

Dataset Used:

psych-ther-ann-rep-csv-2024-25-psych-meds-2

SUBMITTED BY:

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DATA ANALYTICS INTERN

SUBMITTED TO:

ZENO TALENT

Introduction

Prescribing Trends Analysis focuses on studying the patterns of antidepressant and anxiolytic medication prescriptions across different Clinical Commissioning Groups (CCGs) using Open Prescribing data. The objective of this analysis is to identify how prescribing behavior changes over time, compare prescription volumes between CCGs, and observe overall mental health medication trends.

Using **Python**, **Pandas**, and **Matplotlib** in **Google Colab**, the dataset is cleaned, processed, and visualized through trend charts. The analysis helps healthcare researchers and NHS stakeholders understand prescribing demand, monitor treatment practices, and support data-driven healthcare planning.

Technologies Used

- **Python** → data analysis programming language
- **Pandas** → data cleaning, aggregation, trend analysis
- **Google Colab Notebook** → cloud-based notebook environment for running Python code
- **Matplotlib**

Prescribing trends analysis

Prescribing Trend Line Chart

The line chart displays yearly prescribing patterns for antidepressant and anxiolytic medications. It helps identify:

- Increasing or decreasing prescription trends
- Changes in mental health medication demand
- Long-term prescribing behavior
- Comparison between antidepressant and anxiolytic usage

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Commands Code Text Run all

```
[1] ✓ import pandas as pd
import matplotlib.pyplot as plt

[2] ✓ file_path = '/content/psych-ther-ann-rep-csv-2024-25-psych-meds-2.csv'

[3] ✓ df = pd.read_csv(file_path)

[4] ✓ print("Dataset Preview:")
print(df.head())
```

Dataset Preview:

	OrgType	OrgCode	OrgName	VariableType	
0	Commissioning Region	UNKNOWN	UNKNOWN	Psychotropic Medication Status	
1	Commissioning Region	UNKNOWN	UNKNOWN	Psychotropic Medication Status	
2	Commissioning Region	UNKNOWN	UNKNOWN	Psychotropic Medication Status	
3	Commissioning Region	UNKNOWN	UNKNOWN	Psychotropic Medication Status	
4	Commissioning Region	UNKNOWN	UNKNOWN	Psychotropic Medication Status	

	VariableA	VariableB	
0	Status at start - Total	NaN	
1	Status at start - Prescribed but not taking	NaN	
2	Status at start - Prescribed and taking	NaN	
3	Status at start - Not Prescribed	NaN	
4	Status at start - Not known/Not stated/Invalid	NaN	

	Count_StatusEndTotal	Count_StatusEndPrescribedButNotTaking	
0	785	15	
1	18	*	
2	278	5	
3	378	5	
4	55	*	

	Count_StatusEndPrescribedAndTaking	Count_StatusEndNotPrescribed	
0	268	318	
1	5	*	

Variables Terminal

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Commands Code Text Run all

```
[1] ✓ print("\nDataset Information:")
print(df.info())
```

Dataset Information:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 995 entries, 0 to 994
Data columns (total 11 columns):
# Column                                Non-Null Count  Dtype
---  ---                                -
0 OrgType                                995 non-null   object
1 OrgCode                                995 non-null   object
2 OrgName                                995 non-null   object
3 VariableType                            995 non-null   object
4 VariableA                                995 non-null   object
5 VariableB                                0 non-null     float64
6 Count_StatusEndTotal                    995 non-null   object
7 Count_StatusEndPrescribedButNotTaking    995 non-null   object
8 Count_StatusEndPrescribedAndTaking        995 non-null   object
9 Count_StatusEndNotPrescribed              995 non-null   object
10 Count_StatusEndNotStatedNotKnownInvalid  995 non-null   object
dtypes: float64(1), object(10)
memory usage: 85.6+ KB
None
```

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Colab interface showing dataset information and missing values analysis.

Dataset Information:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 995 entries, 0 to 994
Data columns (total 11 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0    OrgType                                     995 non-null    object
1    OrgCode                                    995 non-null    object
2    OrgName                                    995 non-null    object
3    VariableType                               995 non-null    object
4    VariableA                                  995 non-null    object
5    VariableB                                  995 non-null    object
6    Count_StatusEndTotal                       995 non-null    float64
7    Count_StatusEndPrescribedButNotTaking      995 non-null    object
8    Count_StatusEndPrescribedAndTaking         995 non-null    object
9    Count_StatusEndNotPrescribed               995 non-null    object
10   Count_StatusEndNotStatedNotKnownInvalid    995 non-null    object
dtypes: float64(1), object(10)
memory usage: 85.4+ KB
None
```

Missing Values:

```
OrgType          0
OrgCode          0
OrgName          0
VariableType     0
VariableA        0
VariableB        0
Count_StatusEndTotal  995
Count_StatusEndPrescribedButNotTaking  0
Count_StatusEndPrescribedAndTaking  0
Count_StatusEndNotPrescribed  0
Count_StatusEndNotStatedNotKnownInvalid  0
dtype: int64
```

Colab interface showing column name normalization and column availability checks.

df.columns = df.columns.str.strip().str.lower().str.replace(' ', '_')

Column Names:

```
Index(['orgtype', 'orgcode', 'orgname', 'variabletype', 'variablea',
       'variableb', 'count_statusendtotal',
       'count_statusendprescribedbutnottaking',
       'count_statusendprescribedandtaking', 'count_statusendnotprescribed',
       'count_statusendnotstatednotknowninvalid'],
      dtype='object')
```

Unavailable Columns:

```
orgtype
orgcode
orgname
variabletype
variablea
variableb
count_statusendtotal
count_statusendprescribedbutnottaking
count_statusendprescribedandtaking
count_statusendnotprescribed
count_statusendnotstatednotknowninvalid
```

required_columns = ['cog_name', 'antidepressant_items', 'antipsychotic_items']

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```

[132] ✓ 0 required_columns = [
      'cpg_name',
      'antidepressant_items',
      'anxiolytic_items'
]

[133] ✓ 0 required_columns = [col for col in required_columns if col in df.columns]

[134] ✓ 0 trend_df = df[required_columns]
      print("\nSelected Data:")
      print(trend_df.head())

Selected Data:
  cpg_name  antidepressant_items  anxiolytic_items
0  UNKNOWN                    200.0             15.0
1  UNKNOWN                      5.0              0.0
2  UNKNOWN                    185.0              5.0
3  UNKNOWN                     65.0              5.0
4  UNKNOWN                      5.0              0.0

[135] ✓ 0 df = df.rename(columns={'orgname': 'cpg_name'})

[136] ✓ 0 print(df.columns)

Index(['orgtype', 'orgcode', 'cpg_name', 'variabletype', 'variablea',
      'variableb', 'count_statusendtotal', 'anxiolytic_items',
      'antidepressant_items', 'count_statusendnotprescribed',
      'count_statusendnotstatednotknowninvalid'],
      dtype='object')

```

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```

[136] ✓ 0 df = df.rename(columns=[
      'count_statusendprescribedandtaking': 'antidepressant_items',
      'count_statusendprescribedbutnottaking': 'anxiolytic_items'
])

# Convert 'antidepressant_items' and 'anxiolytic_items' to numeric, handling non-numeric values
df['antidepressant_items'] = pd.to_numeric(df['antidepressant_items'], errors='coerce').fillna(0)
df['anxiolytic_items'] = pd.to_numeric(df['anxiolytic_items'], errors='coerce').fillna(0)

# Define the required columns based on the new names
required_columns = [
      'cpg_name',
      'antidepressant_items',
      'anxiolytic_items'
]

# Create trend_df using the (now valid) required_columns
trend_df = df[required_columns]

print("\nSelected Data (trend_df preview):")
print(trend_df.head())
print("\nData types in trend_df:")
print(trend_df.info())

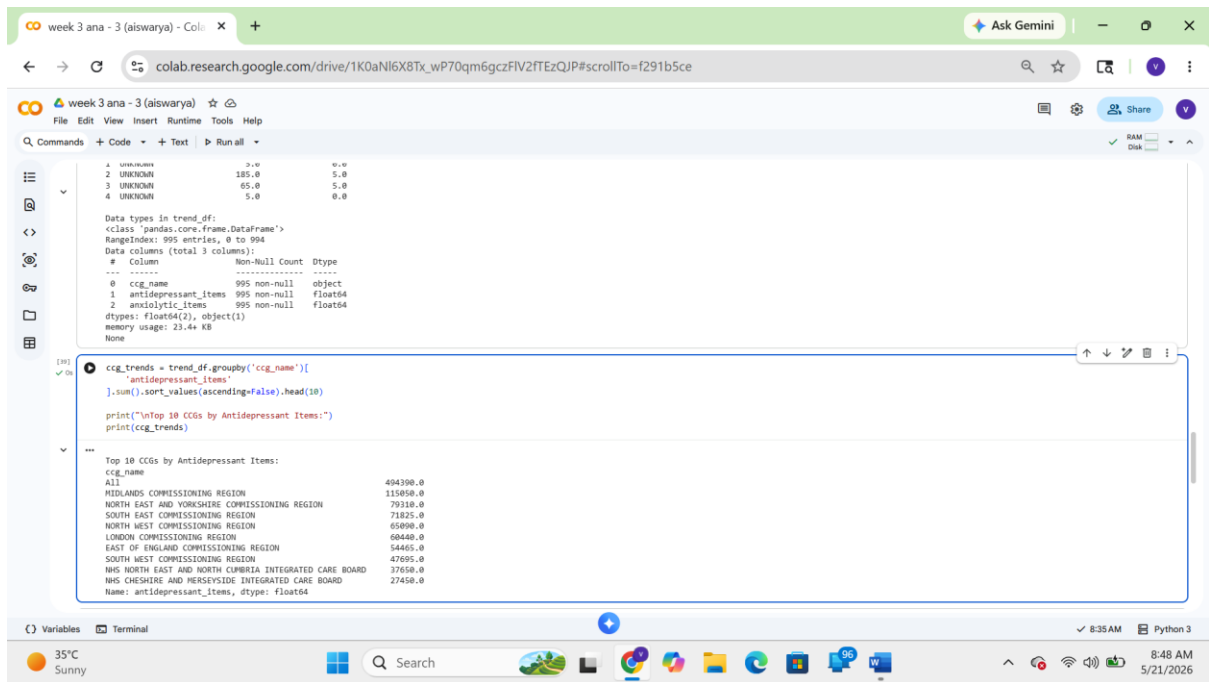
Selected Data (trend_df preview):
  cpg_name  antidepressant_items  anxiolytic_items
0  UNKNOWN                    200.0             15.0
1  UNKNOWN                      5.0              0.0
2  UNKNOWN                    185.0              5.0
3  UNKNOWN                     65.0              5.0
4  UNKNOWN                      5.0              0.0

Data types in trend_df:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 995 entries, 0 to 994
Data columns (total 3 columns):
 #   Column              Non-Null Count  Dtype

```

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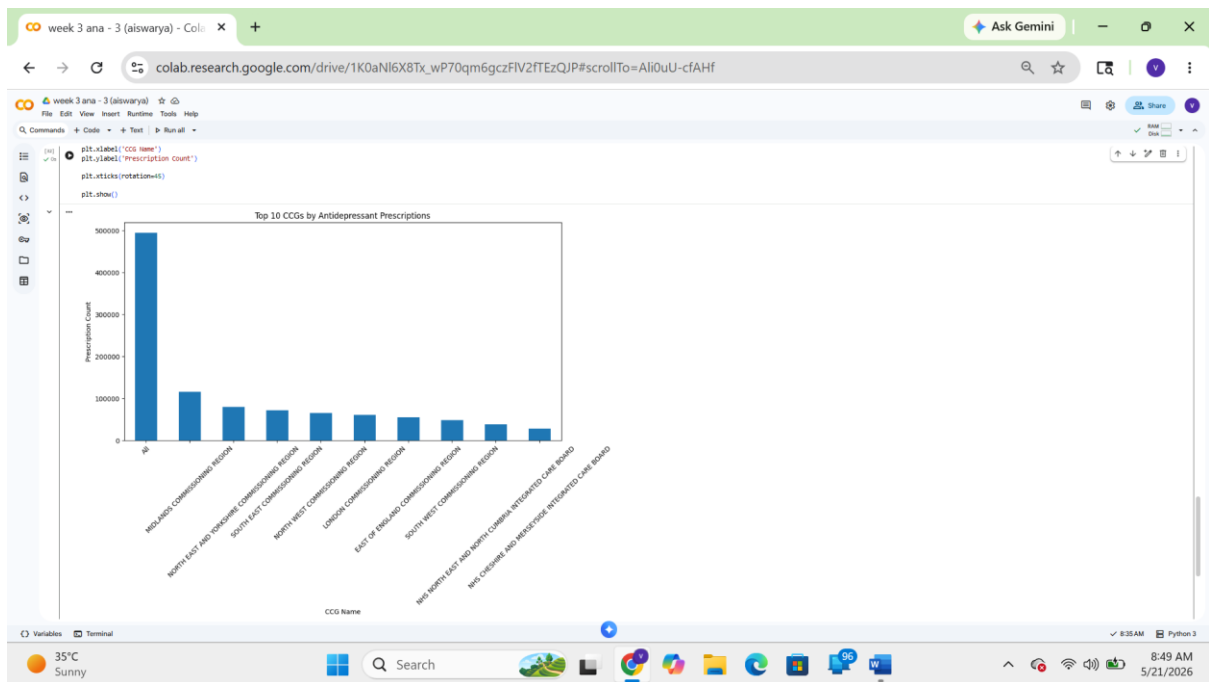
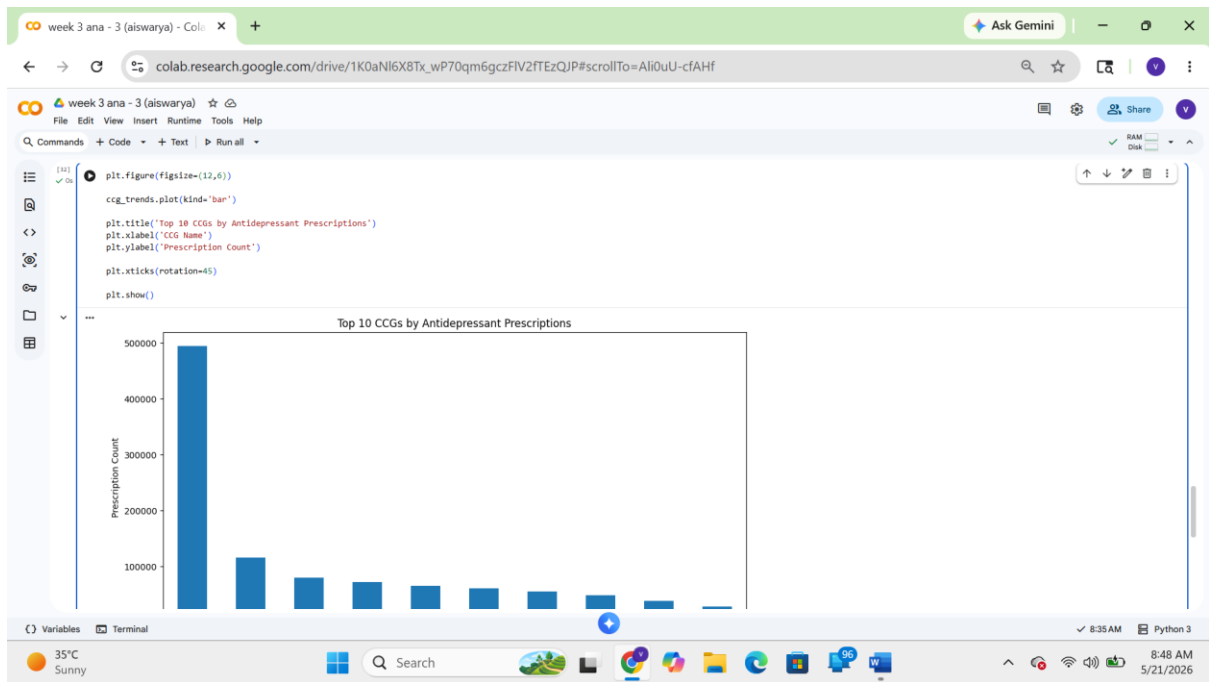


Top 10 CCG Bar Chart

The bar chart compares the top Clinical Commissioning Groups (CCGs) with the highest antidepressant prescription counts.

Insights:

- Identifies regions with high prescribing activity
- Helps compare healthcare demand across CCGs
- Useful for regional healthcare resource planning



Conclusion

The Prescribing Trends Analysis provides valuable insights into antidepressant and anxiolytic prescribing behavior across NHS CCGs. By using Python, Pandas, and Matplotlib in Google Colab, the analysis effectively identifies yearly medication trends and regional prescribing differences.

The visualizations help healthcare professionals and policymakers understand mental health treatment patterns, monitor prescribing growth, and support better healthcare planning. Overall, this analysis demonstrates how data analytics can improve evidence-based decision-making in public health and mental healthcare services.